



School of Electronic Engineering

RFID Side Channel Analysis for IoT Devices Project Portfolio

Katherina Keogh
ID Number: 19423752

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Supervised by Liam Meany

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RFID Side Channel Analysis for IoT Devices

Katherina Keogh
 MEng. Electronic and Computer
 Engineering,
 Dublin City University,
 Dublin, Ireland
 katherina.keogh24@mail.dcu.ie

Abstract— Side channel attacks on RFID devices present a significant threat by enabling attackers to extract secrets through indirect information such as power consumption or electromagnetic (EM) emissions. Existing approaches to RFID security often focus on cryptographic protocol analysis, but may overlook vulnerabilities exposed by physical layer leakage. This paper applies side channel analysis techniques to MIFARE Classic cards using a Proxmark3 reader, an RTB2002 oscilloscope, and probes for both power and EM emissions. Controlled authentication sequences with known keys and data were repeated and measured, generating CSV samples for power and EM signals across multiple RFID memory blocks. These datasets were analysed using leakage modelling and side channel evaluation metrics to assess the possibility of key retrieval. The findings demonstrate the practical risks associated with existing RFID systems.

I. INTRODUCTION

MIFARE Classic cards represent one of the most widely used contactless smart card technologies, used in transport ticketing, building access control, and payment systems worldwide. These cards use the Crypto1 stream cipher to secure mutual authentication and data transmission. Although Crypto1 has been broken through various cryptanalytic techniques, including brute force and algebraic attacks, the potential for additional key related information leakage by physical side channels lacks research [1].

Side channel attacks exploit unintended information leakage through physical factors such as power consumption, electromagnetic (EM) emissions, timing variations, or acoustic emissions. Unlike traditional cryptanalytic attacks that target mathematical weaknesses in algorithms, side channel attacks focus on implementations which are dependent on physical characteristics that may reveal secret keys through statistical analysis of measurement data.

Previous research has demonstrated successful side channel attacks against various cryptographic

implementations, particularly through Differential Power Analysis (DPA) and Correlation Power Analysis (CPA) techniques [2]. These methods exploit correlations between measured power consumption and intermediate cryptographic values, often using leakage models such as Hamming weight or Hamming distance of processed data.

This paper examines whether MIFARE Classic authentication commands affect power consumption and EM emissions differently based on the Crypto1 key used. The power usage and EM emissions were measured and recorded, and statistical analysis was used to evaluate the potential for identifying keys through side channel analysis.

II. PRIOR WORK

Side channel analysis has been widely applied to cryptographic implementations, including smart cards and RFID systems. One study applied machine learning methods such as Support Vector Machines and Rotation Forests to side channel traces, showing that, when combined with effective feature selection, they can outperform traditional template attacks in noisy environments [3]. This demonstrates the value of statistical and machine learning approaches in extracting meaningful information from leakage traces.

Another study focused on RFID smart cards implementing 3DES encryption, demonstrating successful power analysis attacks despite built-in countermeasures [4]. By targeting intermediate rounds of encryption and introducing new distinguishers, the authors were able to significantly reduce key search complexity, ultimately recovering full 168 bit keys. Their findings highlight the vulnerabilities of RFID implementations even when protection schemes are applied.

While much of the existing work focuses on stronger ciphers such as AES and 3DES, there has been comparatively little investigation of the MIFARE Classic's proprietary Crypto1 cipher. This creates a gap in the literature, as it remains unclear whether measurable power and electromagnetic leakage during Crypto1 authentication can be used for key discrimination. This project aims to address that gap by applying side channel analysis techniques to MIFARE Classic authentication traces.

III. EXPERIMENT SETUP AND IMPLEMENTATION

This paper investigates side channel leakage from MIFARE Classic RFID cards by measuring power consumption and EM emissions during authentication. The aim is to investigate whether the secret key used for encryption or block data processed by the Crypto1 stream cipher produces measurable physical signatures exploitable in practical attacks.

A. Experimental Hardware Setup

The hardware used in this experiment consisted of a Proxmark3 device, a Rohde & Schwarz RTB2002 digital oscilloscope, a RaspberryPi and a MIFARE Classic 1K RFID card.

The Proxmark3 was used to communicate directly with the RFID card and issue commands such as key authentication and block writing. The Iceman firmware was running on the Proxmark3. The Iceman repo is recognised for its extensive features, active development, and broad compatibility with various RFID standards [5], which allow advanced commands and scripting. The supportive firmware documentation made it especially effective for tasks such as MIFARE Classic card manipulation, providing reliability and flexibility. The Proxmark3 also acted as the trigger source for synchronising oscilloscope captures.

A RIGOL Passive Probe PVP3150 was connected to the Proxmark to capture the power used when executing the commands.

A high sensitivity, near field EM probe, NAE Hprobe, was used to capture side channel emissions during card communication.

The RTB2002 oscilloscope was used to record traces from these two analog channels, measuring EM and power related signals during MIFARE operations. The horizontal scale was configured to cover the full authentication exchange, , and the

vertical scaling was adjusted individually per channel to maximise signal to noise ratio without saturation.

The MIFARE Classic 1K card served as the target of the analysis. It was placed directly on the Proxmark, which was attached to a sensing probe and the EM probe was placed directly on top of the MIFARE card during trace collection to ensure consistency.

A Raspberry Pi 3 running Raspberry Pi OS was used to power the Proxmark3 via its USB port and to orchestrate authentication sequences. The Raspberry Pi 3 was used to interface with the Proxmark3 and manage data collection due to its Linux based operating system and small size which is ideal for integrating both the Proxmark3 and oscilloscope into a compact, self contained measurement platform. The Pi issued commands over USB to the Proxmark3 and mounted the RTB2002's USB stick storage so that the CSV trace files could be transferred to the Pi's filesystem before being uploaded to Google Drive.

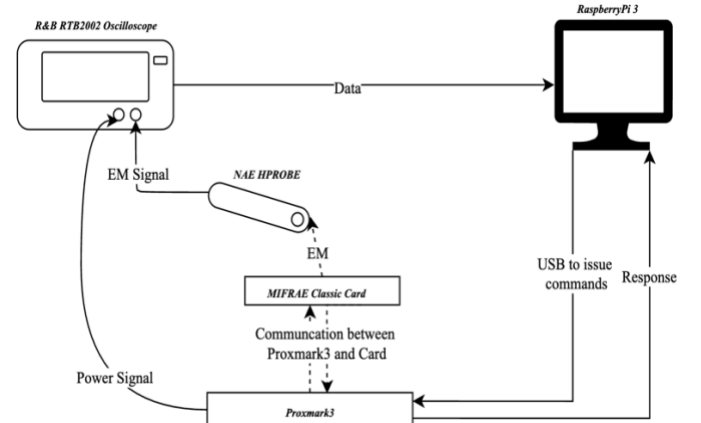


Fig 1: Block Diagram of Hardware Setup

B. MIFARE Classic Setup

The MIFARE Classic card is organised into sectors, each containing four data blocks. The last block in each sector is known as the sector trailer and stores two 6 byte cryptographic keys, Key A and Key B. These keys are represented in hexadecimal format and consist of 12 hexadecimal characters. The keys are used to control access to the data within the sector. [6]

The Proxmark3 was used to communicate with the MIFARE Classic card. Initial identification of the card type and basic properties was performed using the 'hf search' and 'hf mf info' commands, which

revealed the UID, ATQA, SAK, and detected keys [7]. These commands provided confirmation of the card type and manufacturer, ensuring that it was compatible with Crypto1 based authentication. Before running the authentication command, individual authentication tests, ‘hf mf auth’ and read operations were performed to validate known keys and confirm sector access.

Three distinct Crypto1 keys were programmed into different sectors of the MIFARE Classic card to enable comparative analysis. The keys were selected at random.

Block	Key
Block 1	434A9734E087
Block 2	10203040506
Block 3	A0A1A2A3A4A5

Table 1: Block and Key combinations

The Proxmark3 ‘hf mf wrbl’ command was used to program sector trailer blocks with the respective keys, followed by verification using ‘hf mf chk’ commands [7].

The ‘hf mf chk’ command is a key checking function used to test a supplied key against a target block on a MIFARE Classic card [7]. When executed, the command triggers a full authentication sequence using the Crypto1 cipher, making it highly relevant for side channel analysis. This sequence involves the exchange of authentication nonces and the generation of cryptographic keystreams, during which measurable power and EM emissions occur.

The command was chosen because it creates a consistent authentication process with few extra steps, which produce clear and comparable data for analysis. Its predictable nature and direct use of the target key make it a good choice for checking if different keys show different, noticeable patterns in side channel measurements.

Please see Appendix H for Proxmark3 Command Line Interface output screenshots illustrating these commands.

C. Data Collection

To gather consistent side channel data, a list of repeatable measurement steps was created. For each test, a single MIFARE Classic 1K card was placed directly on the Proxmark with the EM probe placed on top of the card, which was connected to the oscilloscope. Each test sequence involved sending the same command multiple times to the same block,

with the hardware setup remaining in the same physical position throughout the experiments to minimise any variations in the data.

The authentication sent from the Proxmark was repeated 15 times for each block and key combination. Each command triggered a waveform capture on the oscilloscope, which recorded power, EM and timing data. The oscilloscope was configured to begin capturing as soon as the command was issued, using digital triggering based on signal activity to align each trace.

Waveforms were saved in CSV format with filenames clearly indicating the key, and repetition number. This ensured proper trace tracking and data labelling during analysis. All captures were carried out using the same sampling rate, voltage scale, and time base settings to ensure consistency across the entire dataset. There was a total of 45 CSV files.

This list of steps allowed for fair comparisons between different keys and blocks, making it possible to evaluate whether observable variations in power or EM patterns could be linked to the specific key used.

D. Data Processing and Visualisation

All of the data preprocessing, visualisation, and statistical analysis were conducted using Google Colab, a cloud based Jupyter notebook environment. Google Colab was chosen for its accessibility, preconfigured Python libraries, and integration with Google Drive for storing the CSV files [8]. Google Colab provided access to essential libraries, including NumPy for numerical operations and array handling of waveform data, Pandas for data manipulation and organisation of trace collections, Matplotlib for generating plots of power and EM traces, and SciPy for statistical analysis functions, including t-tests and correlation calculations. The cloud based environment also eliminated the need for local software installation. The oscilloscope waveform files were uploaded to Google Drive and were accessed directly within the Colab notebooks, allowing for easy data import and preprocessing techniques. All statistical computations, including Cohen’s d effect size calculations and Pearson correlation analysis between key Hamming weight and EM peak values, were performed using standardised Python functions within the Colab environment.

The data sets were first trimmed to begin at the authentication command's time zero, ensuring the analyses focused on the Crypto1 processing window and eliminated noise not relevant to the command. The 15 samples for each key were then combined and the average was taken. The selected preprocessing techniques are designed to ensure that comparable metrics were obtained for subsequent t-tests, effect-size calculations, and correlation analysis.

IV. RESULTS OBTAINED

A. Trace Characteristics

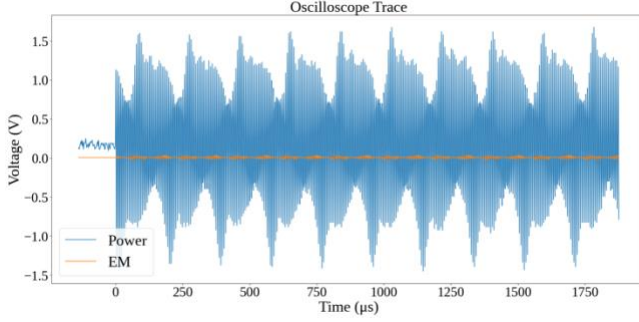


Fig 2:

The snippet of an oscilloscope trace in Fig 2 presents the simultaneous measurement of power and EM signals collected during a typical RFID transaction. The power signal displays rhythmic oscillations spanning voltage values from around -1.5V to 1.5V, reflecting the alternating nature of the underlying RFID communication and energy exchanges between reader and tag. The EM signal, in orange, captured in parallel but displaying significantly lower amplitude, close to the zero volt baseline, represents the EM leakage associated with device activity. While it is less noticeable, this signal holds important information for security analysis. Its slight changes can expose vulnerabilities that attackers could exploit.

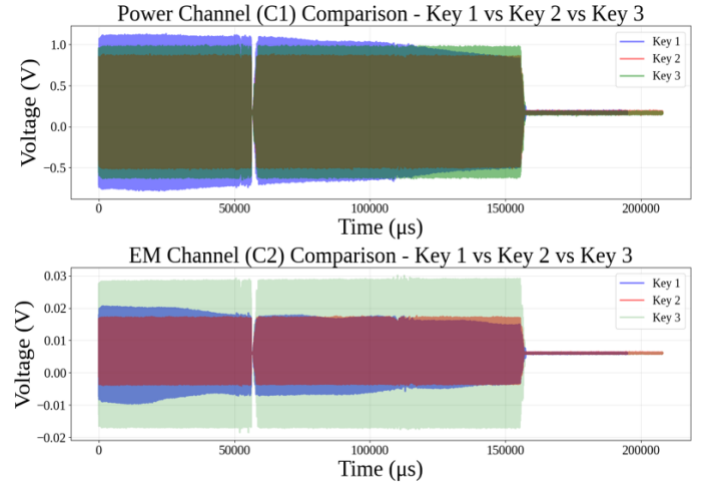


Fig 3: Overlay of averaged power (C1) and EM (C2) traces for all three keys during MIFARE Classic authentication

Figure 3 shows the averaged side channel traces for all three keys and blocks. Both power and EM measurements show identical timing structure, confirming that all traces successfully captured the same 'hf mf chk' authentication protocol.

For each key, Channel 1 displays the average power profile, showing how voltage fluctuates from negative to positive values throughout the transaction, typically ranging between approximately $-0.75V$ to $1.0V$ for key 1, $-0.4V$ to $0.8V$ for key 2, and $-0.5V$ to $1.0V$ for key 3. Channel 2 then displays the corresponding EM leakage, with shifts in voltage observed in the range of $-0.01V$ to $0.02V$ for keys 1 and 2, and a slightly wider spread of $-0.02V$ to $0.03V$ for key 3.

The time axis shows these processes happening quickly, emphasizing the repeated yet complex interactions in RFID communications.

A comparison across all three keys reveals similar overall signal structure and timing, with two major peaks and a pronounced dip observed around the middle of each trace (approximately $50,000 \mu s$). These features align with critical phases of RFID protocol execution, likely corresponding to authentication or data transfer events. The consistency in shape across all keys suggests a uniform operational process, while minor differences in amplitude and spread may reflect variations in the cryptographic computations or the underlying key dependent operations.

By averaging the traces for each block/key, noise is reduced and recurring features are accentuated,

allowing for comparative analysis for potential side channel leakage. The subtle distinctions in EM channel amplitudes, especially in key 3, could be significant when probing for key dependent variations, highlighting regions of interest for further security analysis. Overall, these visualisations are useful for identifying protocol phases, assessing process uniformity, and pinpointing windows where sensitive operations occur.

From these traces, it was determined that the timings were almost identical, but there was a visible difference in amplitude.

B. Statistical Analysis

The statistical analysis in this study compares signal features between different keys using the t -statistic, p -value, and Cohen's d effect size. The t -statistic tests whether the means of two independent groups differ significantly, while the p -value indicates the likelihood that any observed difference is due to chance [9]. The formulas for the t -statistic, degrees of freedom, and p -value are provided in Appendix G. These metrics are essential for side channel analysis, as they reveal whether variations in power and EM signals are statistically significant, suggesting key dependent leakage.

The statistical comparison between key pairs was quantified using Cohen's d , a metric for measuring effect size which is the standardised difference between two means. This standardisation helps show not just whether a difference exists, but how large that difference is.

Table 1 summarises the statistical comparisons of side channel features between each pair of keys. The mean and peak amplitudes in power (C1) and EM (C2) channels over a 0 – 110 000 μ s window were analysed.

Pair	Channel	Metric	T-statistic	P-value	Cohen's d
Key1 vs Key2	C1	Mean	1.785	0.083	0.5913
Key1 vs Key2	C2	Mean	-1.927	0.0625	-0.6343
Key1 vs Key2	C1	Peak	5.21	0	1.8591
Key1 vs Key2	C2	Peak	5.737	0	1.9045
Key1 vs Key3	C1	Mean	5.611	0	1.8204
Key1 vs Key3	C2	Mean	0.225	0.823	0.0731
Key1 vs Key3	C1	Peak	14.979	0	4.8598
Key1 vs Key3	C2	Peak	-4.699	0.0001	-1.5246
Key2 vs Key3	C1	Mean	4.222	0.0002	1.4451
Key2 vs Key3	C2	Mean	2.401	0.0218	0.8019
Key2 vs Key3	C1	Peak	2.511	0.0212	0.9125
Key2 vs Key3	C2	Peak	-6.418	0	-2.0088

Table 2: Statistical Test Results for Power (C1) and EM (C2) Channels

Peak amplitudes differ significantly for every key pair in both channels, with Cohen's d ranging from 0.886 to 4.737. Mean amplitudes also show significant differences in most comparisons, particularly for the power channel. These results confirm that both power and electromagnetic side channels leak key dependent information, and that peak based features show the most significant difference.

C. Hamming Weight Correlation Analysis

Key	HW
Key 1	21
Key 2	9
Key 3	19

Table 3: Hamming Weight of Keys per Key

Given the limitations of traditional statistical tests, CPA techniques were used to investigate whether side channel amplitudes correlate with the Hamming weight of the processed keys. Hamming weight, which represents the number of '1' bits in a binary key, is an indicator of how internal data operations may affect physical leakage in cryptographic devices.

For each individual signal trace, the peak voltage was extracted within the primary processing window (0 – 150,000 μ s) from both power and EM channels. These peak values were then correlated with the corresponding key Hamming weights using Pearson's correlation coefficient. The formulas for the Pearson's correlation coefficient is provided in the appendices.

This correlation measures the strength of the linear relationship between key Hamming weights and side channel measurements. A strong positive correlation provides direct evidence that the physical signals leak key dependent information, confirming the vulnerability of the system to side channel attacks. This analysis strengthens the idea that key dependent leakage is beyond what statistical significance can demonstrate on its own.

D. Leakage Model Validation

The scatter plots illustrate the correlation between key Hamming weight and peak channel voltage for both power (C1) and EM (C2) measurements. Each dot on the scatter plot represents the data from one

CSV file, showing the signal recorded for a single authentication attempt and the Hamming weight of the key used.

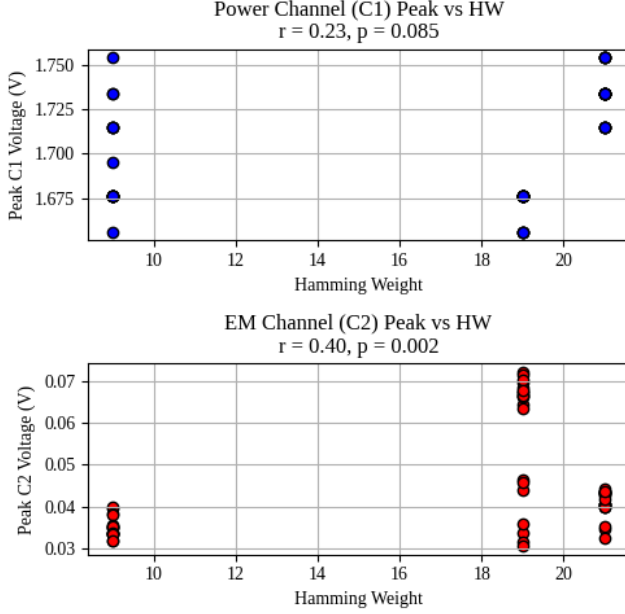


Fig 3: Correlation Between Peak and Hamming Weight For C1 and C2

In the power channel (C1) plot the correlation coefficient $r = 0.23$ and p-value $p = 0.085$ suggest only a weak, statistically nonsignificant relationship between Hamming weight and measured peak voltage. This indicates that, in this dataset, power measurements provide limited information on key dependent leakage.

By contrast, the EM channel (C2) plot shows a notably stronger correlation ($r = 0.40$, $p = 0.002$), which is statistically significant. Here, keys with higher Hamming weights consistently display higher peak EM emissions, as reflected by the cluster of points with both high Hamming weights and elevated voltages.

V. ANALYSIS

A. Interpretation of Statistical Results

The statistical analysis reveals evidence of key dependent side channel leakage in MIFARE Classic authentication. The Cohen's d values, ranging from 0.886 to 4.737, represent effect sizes that exceed the conventional threshold for significant effects ($d \geq 0.8$). These magnitudes suggest that the observed differences between keys are not only statistically

significant but practically meaningful for cryptographic exploitation.

The consistently higher effect sizes observed in EM measurements compared to power analysis suggest that EM emissions provide a more reliable side channel factor for MIFARE Classic cards. This also coincides with previous research demonstrating that EM probes can capture spatially localised leakage that may be obscured in power measurements due to circuit complexity and current distribution.

B. Correlation with Handshake Phases

By aligning filtered traces to the start of the 'hf mf chk' command, distinct waveform segments corresponding to reader challenge, card response and keystream generation are identifiable. Peak EM leakage occurs during keystream generation, when the card's LFSR updates are processed, confirming that physical emissions directly map to the Crypto1 round function. This mapping establishes a causal link between protocol activity and measured leakage, demonstrating that an attacker observing specific handshake phases can target analysis to the most informative time windows.

C. Practical Attack Feasibility

The correlation coefficient of $r = 0.40$ between EM peak amplitude and key Hamming weight, whilst statistically significant ($p = 0.002$), represents a moderate correlation strength. This indicates that approximately 16% of the variance in EM emissions can be due to the key dependent characteristics. In the context of side channel analysis, this level of correlation suggests sufficient leakage for practical exploitation, particularly when combined with advanced attack methodologies.

The observed leakage pattern follows the established Hamming weight model, where cryptographic operations on data with higher bit densities produce correspondingly higher physical emissions. This relationship has been successfully exploited in numerous side channel attacks against various cryptographic systems, including AES, DES, and RSA [2].

D. Implications for Practical Security

Although key recovery was not demonstrated within the scope of this study, the statistical evidence strongly supports the possibility of practical attacks given appropriate resources and methodology

refinement. The combination of large effect sizes, significant correlations, and established leakage models creates a foundation for advanced attack development.

MIFARE Classic cards are commonly used in sensitive areas like building access, public transport, and payment systems. The recent discovery of a security flaw adds a new way for attackers to exploit these cards, alongside existing weaknesses in their encryption system, Crypto1.

E. Limitations and Environmental Factors

The moderate sample size of 15 per key represents a significant limitation for definitive conclusions about attack success rates. Advanced side channel attacks typically require thousands of traces for reliable key recovery, particularly in noisy environments. Larger datasets are likely to show stronger attack success rates.

A further constraint arose from the difficulty in automating the measurement process. Each trace had to be triggered and captured manually, which limited the total number of samples that could be collected within the project timeframe. As a result, the dataset is considerably smaller than that used in comparable published studies.

Environmental noise and measurement variability contributed to signal degradation, as evidenced by the scatter in correlation plots. The conditions in which the experiments were conducted did not replicate the controlled environments typically required for sophisticated side channel attacks, where factors such as temperature stability, electromagnetic interference, and mechanical vibrations are controlled.

VI. CONCLUSION

This study examined whether MIFARE Classic authentication produces measurable power and EM side channel leakage that could reveal information about the secret keys used for Crypto1 encryption. Using a Proxmark3 to trigger repeated authentication commands and an oscilloscope to record both power and EM signals, distinct patterns were observed between keys of differing Hamming weight. Statistical analysis demonstrated a strong correlation between the peak amplitudes of EM signals and the key Hamming weight. This suggests that MIFARE Classic systems might reveal sensitive information

beyond the known weaknesses in the Crypto1 encryption method.

Although the dataset was limited in size, the results suggest a potential for key discrimination or recovery through targeted side channel analysis. Expanding the sample set, applying advanced techniques such as correlation power analysis, and exploring additional leakage models could strengthen these findings. The evidence presented reinforces the importance of considering physical leakage in the security evaluation of contactless smart cards, even for systems already compromised at the algorithmic level.

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APPENDIX A

LITERATURE REVIEW



School of Electronic Engineering

RFID SIDE CHANNEL ANALYSIS FOR IoT DEVICES

Literature Survey

Katherina Keogh
ID Number: 19423752

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MEng in Electronic and Computer Engineering

Supervised by Liam Meany

DECLARATION

I hereby declare that, except where otherwise indicated, this document is entirely my own work and has not been submitted in whole or in part to any other university.

Signed: Katherina Keogh

Date: 19th of Jan 2025

RFID Side Channel Analysis for IoT Devices

Katherina Keogh
MEng. Electronic and Computer
Engineering,
Dublin City University,
Dublin, Ireland
katherina.keogh24@mail.dcu.ie

Abstract—The rapid increase of Radio Frequency Identification (RFID) technology within the Internet of Things (IoT) ecosystem has revolutionised applications ranging from inventory management to healthcare. However, alongside its benefits, RFID systems face critical security vulnerabilities, particularly in the context of side-channel analysis (SCA). This paper investigates the susceptibility of RFID systems to sidechannel attacks, which exploit unintended physical emissions such as power consumption and electromagnetic radiation to extract sensitive information. Supported by prior research and analysis, this work highlights key security challenges, focusing on vulnerabilities in RFID protocols and their implications for privacy and data integrity. The potential of enhancing sidechannel analysis techniques using tools such as the Proxmark 3 proposing firmware modifications to improve its utility in evaluating RFID security is also discussed.

Index Terms— Internet of Things, Proxmark, Radio Frequency Identification, Side Channelling

I. INTRODUCTION

Radio Frequency Identification (RFID) technology has become an important part of the Internet of Things (IoT) ecosystem, enabling communication and identification in applications ranging from supply chain management to healthcare and smart cities [1]. As RFID systems continue to grow, their security and reliability have lacked significant attention, particularly in protecting sensitive data and ensuring operational integrity [2]. Despite advancements in encryption and authentication protocols, RFID systems remain susceptible to various vulnerabilities, including side-channel attacks, which exploit unintended information leaks to compromise security [3]. Side-channel analysis (SCA) is a growing area of concern

in RFID security, using signals such as timing information, power consumption, and electromagnetic emissions to extract confidential data [3]. These non-invasive attacks bypass traditional cryptographic defences, making them hard for RFID systems to detect. Given the increasing deployment of RFID in security-critical applications, understanding and reducing the risks posed by SCA is essential to enhancing IoT device security.

This review explores the RFID technology and side-channel analysis, focusing on identifying vulnerabilities and evaluating the potential of tools like the Proxmark 3 to display their weaknesses.

II. RFID IN IoT

An RFID (Radio Frequency Identification) system is a wireless communication technology for automatically identifying and tracking objects, animals, or people. It consists of three main components: a tag, a reader, and a backend system. The tag, which can be either passive, active, or semiactive, contains a microchip for storing data and an antenna for transmitting that data to a reader. The reader emits radio frequency signals to power and communicate with the tag in the case of passive tags or to receive data from active tags that have their own power source. The backend system processes the data received by the reader and integrates it into applications such as inventory management, access control, or supply chain logistics. RFID systems operate at various frequency ranges, such as low frequency (LF), high frequency (HF), and ultra-high frequency (UHF), depending on the use case. Their

ability to enable contactless communication over short distances makes them popular in numerous industries.[4]

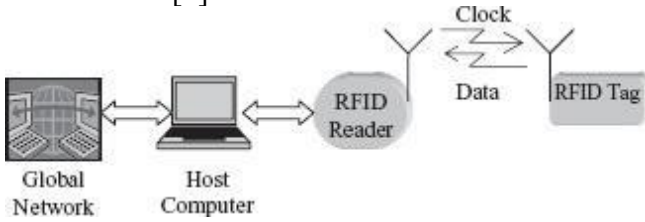


Fig. 1. Block diagram of a typical RFID system.

[4]

In practical applications, RFID has transformed industries by enabling real-time asset tracking, automation, and data collection. Retail giants like Wal-Mart leverage RFID for efficient inventory management [4], while the healthcare industry relies on it for equipment tracking and patient identification. Transportation systems integrate RFID in toll collection and contactless fare cards, and smart cities use the technology to monitor infrastructure and deploy IoT solutions. These implementations highlight RFID's ability to enhance efficiency and precision in complex operations.

Despite these advancements, several underlying aspects of RFID systems can introduce challenges. The reliance on radio frequency signals, while enabling continuous communication, also creates opportunities for interception or disruption of data transmissions. In applications requiring low-cost tags, the computational limitations of passive tags often restrict their ability to implement robust cryptographic protocols [5]. These characteristics make RFID systems attractive targets for malicious actors seeking to exploit authentication or data protection vulnerabilities. As RFID systems become increasingly integrated with IoT networks, the consequences of security weaknesses are amplified, demanding significant research efforts to enhance the resilience of this critical technology.

III. RFID VULNERABILITIES

RFID technology, while revolutionising industries with its data transmission capabilities, faces significant security challenges. Side-channel attacks (SCAs) represent a critical and often underappreciated threat among the numerous attack vectors. SCAs exploit physical characteristics and unintended emissions of RFID devices rather than directly targeting the underlying communication protocols.

RFID systems are vulnerable to side-channel attacks due to their constrained computational resources and reliance on lightweight cryptographic protocols. Unlike traditional attacks such as eavesdropping or cloning, SCAs analyse physical emissions, such as power consumption, electromagnetic radiation, and timing information, generated during the operation of RFID tags or readers [3]. For example, during cryptographic operations like encryption or authentication, variations in power usage or electromagnetic emissions can leak sensitive information, including cryptographic keys. This vulnerability arises because many RFID devices are designed to prioritise cost and efficiency over physical security.

A. Cloning RFID Cards

The study carried out by Jain, Chaudhary, et al. underscores the inherent vulnerabilities in RFID technology, particularly in its current state of implementation. The ability to easily access, alter, or clone data stored on RFID tags reveals significant weaknesses that can compromise privacy and security. The research highlights the susceptibility to denial-of-service attacks, where card data can be maliciously overwritten, further highlighting the fragility of these systems. The experiments performed on a University Identification card and blank cards demonstrated how effortlessly such attacks can be executed, reinforcing the urgent need for advancements in RFID security. [2]

B. Hamming Distance Leakage Model

The study carried out by Xu, Zhu, et al. further emphasises the critical vulnerabilities inherent in RFID and smart card systems when subjected to side-channel attacks. By using Hamming distance leakage and targeting specific rounds of DES encryption, the research highlights how sophisticated combinations of attack methods can bypass existing security measures [3]. This highlights the inadequacy of traditional countermeasures such as "head and tail" protection, which fail to prevent attackers from exploiting subtle leaks in the cryptographic process. These results not only demonstrate the possibility of side-channel attacks but also point to the need for enhanced defensive strategies to reduce these risks and ensure the integrity of sensitive data in RFID systems.

C. Reverse Engineering

The study carried out by Fraga-Lamas and FernándezCaramés further emphasises the importance of securing RFID systems. In this study, the authors reverse-engineered the communication protocol of an RFID public transportation card. They demonstrated how an attacker could manipulate the system to gain unauthorised access and clone the RFID card. This research highlights the vulnerability of RFID in realworld scenarios, particularly in public-facing applications such as transportation, where attackers can exploit communication protocol weaknesses to compromise the system.[6]

D. Machine Learning-Based Side-Channel Attacks Recent advancements in side-channel analysis have introduced machine learning (ML) as a state-of-the-art approach for attacking cryptographic systems. These techniques use ML algorithms to model and exploit physical leakage, such as power consumption or electromagnetic radiation, during sensitive computations. Unlike traditional methods, machine learning-based side-channel attacks are especially powerful in scenarios where only a limited number of profiling traces are available. By combining machine learning methods with side-channel analysis, researchers have identified new ways to bypass the constraints of classical approaches like template attacks (TAs) [7].

To validate these findings, researchers conducted experiments using a novel metric called the Data Confusion Factor (DCF), which measures the success of machine learning models in profiling and attack phases. These experiments showed that ML-based side-channel attacks achieved higher success rates than TAs, especially in cases involving high-dimensional traces with many irrelevant samples. The use of dimensionality reduction techniques, such as principal component analysis (PCA), has further enhanced the efficiency of ML-based attacks by focusing only on the most informative features of the leakage traces [7].

IV. SIDE-CHANNEL ANALYSIS ON RFID RESEARCH

Side-channel analysis (SCA) is a powerful tool for evaluating vulnerabilities in RFID systems. SCA exploits physical information leakage, such as electromagnetic (EM) emissions, power consumption, or timing discrepancies, to infer sensitive data without directly attacking

cryptographic algorithms [3]. Side-channel attacks reveal critical weaknesses in RFID systems by analysing physical manifestations of computations. Differential power analysis (DPA), for instance, can extract cryptographic keys by comparing power consumption across multiple operations, while timing attacks deduce sensitive data based on processing delays. Electromagnetic attacks are particularly relevant to RFID due to the wireless nature of its communication, allowing EM emissions to be intercepted and analysed. [8] The Proxmark is a versatile, open-source tool designed for the analysis, manipulation, and emulation of RFID systems [9]. The device is widely used in security research, enabling tasks such as reading and writing RFID tags, capturing communication data, and performing advanced attacks like cloning, eavesdropping, and replay attacks. With its customisable firmware and powerful hardware capabilities, the Proxmark is a critical platform for investigating vulnerabilities in RFID systems and testing potential countermeasures. To maximise Proxmark's efficacy for SCA, several firmware modifications can be made.

A. High-Resolution Signal Capture

The current functionality of the Proxmark device captures communication signals from RFID systems, but it may lack the granularity required for detailed side-channel analysis. As stated, side-channel attacks often rely on detecting subtle variations in physical signals such as EM emissions or power consumption, meaning these variations can reveal sensitive data, including cryptographic keys or sensitive information being processed by RFID devices.

The firmware could be enhanced to address this limitation and support high-resolution sampling and data storage. Modifications can allow for a more precise analysis of the physical characteristics of RFID communication, improving the ability to detect leaks in physical emissions. For example, the *client/src/cmdhf.c* [10] file, which handles high-frequency commands, could be modified to introduce a high-resolution capture mode. By enhancing this section of the firmware, the Proxmark would be better equipped to collect and store detailed EM data, enabling fine-grained analysis of physical leakage. This capability would facilitate more accurate timing, power, and

electromagnetic analysis, which are crucial for identifying vulnerabilities in RFID systems.

B. Enhanced Timing Analysis Tools

Timing analysis is a common technique in side-channel analysis, relying on precise measurements of processing delays and computation timings. Many side-channel attacks, such as timing attacks, exploit these processing delays to infer sensitive information, such as cryptographic keys or data patterns. These delays are often generated during cryptographic operations in RFID tags or readers.

Upgrading the Proxmark firmware to log timestamps with higher accuracy could significantly improve the reliability of timing-based attack experiments. For example, the file `client/src/cmdhfl4a.c` handles high-frequency ISO14443A commands, which are often used in RFID communication protocols [10]. By modifying this file to increase timestamp resolution, researchers could accurately capture the precise timing of events within the system. This would allow researchers to observe even small delays, which could reveal vulnerabilities in cryptographic operations or authentication protocols.

With this enhancement, researchers could detect variations in processing times that would otherwise go unnoticed with lower-resolution timestamp logging. These subtle variations can be exploited in timing-based side-channel attacks to obtain cryptographic keys or to reveal patterns in data transmission. For example, side-channel analysis could reveal information such as whether a tag is performing encryption, the timing of key generation, or whether certain cryptographic steps are skipped during processing.

Through these proposed firmware modifications, the Proxmark would provide a secure platform for conducting more effective and detailed timing-based side-channel analyses in RFID systems. These tools would not only validate existing research but also explore vulnerabilities that could be exploited in practical scenarios.

V. PRELIMINARY WORK

To establish a functional environment for conducting the research, preliminary setup steps were carried out on a macOS machine. The essential development tools, including Apple's XCode

Command Line Tools, which provide functions like git and make files for building software, had already been previously installed and set up. The Proxmark 3 Iceman firmware repository was then cloned from GitHub using git, enabling access to the latest firmware updates and modifications. Homebrew, a widely-used package manager for macOS, was used to install dependencies such as gcc, libusb, and cmake, ensuring the Proxmark client and firmware could be compiled and executed seamlessly.

The next step will involve downloading the Proxmark client software and following detailed instructions to flash the firmware onto the Proxmark 3 hardware. This will require connecting the device to the Mac via USB, ensuring compatibility with the device's serial drivers. Specific configuration files within the firmware repository, such as `cmdhf.c` and `cmdhfl4a.c`, were examined to identify potential areas for customisation. Testing the setup will involve running basic Proxmark commands to verify communication between the software and hardware.

VI. CONCLUSION

The security of RFID systems remains a critical concern as these technologies become increasingly essential in the Internet of Things (IoT) ecosystem. This paper has examined the vulnerabilities of RFID systems, with a focus on their susceptibility to side-channel attacks (SCAs). By exploiting physical emissions such as electromagnetic emissions, timing variations, and power consumption, SCAs bypass traditional cryptographic protections, exposing sensitive data to malicious actors. These attacks are particularly concerning given the constrained resources of RFID tags, which often prevent the implementation of secure security measures.

Prior research has demonstrated the possibility and effectiveness of SCAs in exposing weaknesses in RFID systems. Studies such as those by Jain et al. and Xu et al. have highlighted the ability of attackers to extract cryptographic keys and manipulate data through methods like Hamming distance leakage analysis and differential power analysis. These findings underline the inadequacy of existing countermeasures and the urgent need for advanced security strategies to protect RFID systems.

The use of tools like the Proxmark 3 offers significant potential in identifying and reducing vulnerabilities. The ability to capture high-resolution

signals and conduct detailed timing analysis makes the Proxmark an invaluable resource for evaluating the security of RFID systems. Proposed firmware enhancements, such as increasing signal capture resolution and improving timing precision, would further strengthen its side-channel research and testing capabilities.

As RFID continues to integrate with critical applications in supply chain management, healthcare, transportation, and smart cities, ensuring the security and resilience of this technology is increasingly important. This research highlights the need for continued research in defensive measures, such as side-channel-resistant cryptographic protocols and improved hardware designs, to further reduce risks. Addressing these challenges is essential to maintaining and protecting sensitive data.

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APPENDIX B

PROJECT PLAN

Side Channel Analysis on RFID Devices

RESEARCH QUESTION

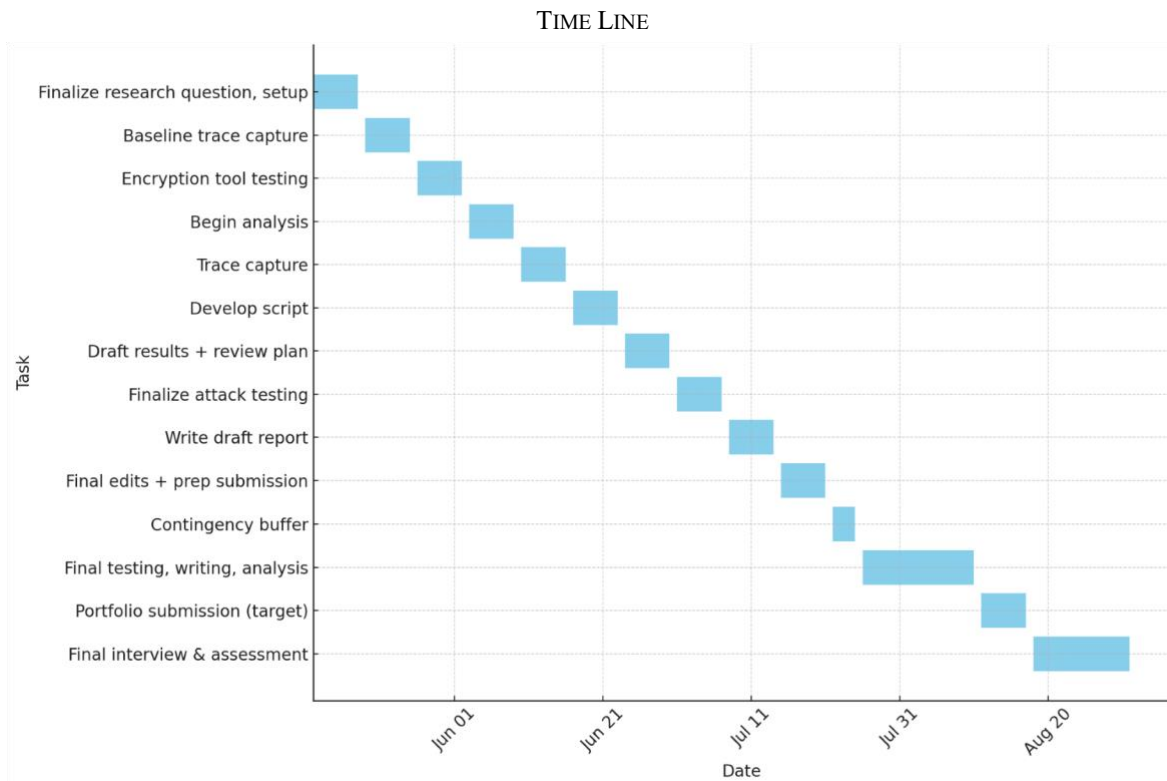
“Can power analysis attacks extract cryptographic information from RFID cards using a Proxmark3, an oscilloscope, and an electromagnetic probe?”

PROJECT SCOPE

- Electromagnetic power analysis of RFID communication.
- Use of the Proxmark3 to trigger and control RFID transactions.
- Integration of an oscilloscope and NAE Hprobe 15 to capture electromagnetic emissions.
- Use of tools for trace capture and analysis.
- Investigation of leakage from RFID cards.
- Manual implementation of encryption for test inputs.
- Side-channel data visualisation and statistical correlation analysis.
- Development of scripts or tools to automate trace alignment and analysis.

DESIGN APPROACH

1. Setup Design:
 - 1.1. Use the Proxmark3 to initiate RFID authentication or read/write operations.
 - 1.2. Position the NAE Hprobe 15 near the RFID card antenna to capture emissions.
 - 1.3. Use an oscilloscope connected to a Mac to record power traces for each interaction.
2. Experimentation:
 - 2.1. Perform repeated RFID operations with fixed and varying keys.
 - 2.2. Record and label corresponding power traces.
 - 2.3. Inject known encrypted values to study electromagnetic changes.
3. Analysis:
 - 3.1. Use side channel techniques to identify key leakage.
 - 3.2. Compare the leakage of weak cards..
 - 3.3. Evaluate noise, signal alignment, and possible countermeasures.
4. Validation:
 - 4.1. Demonstrate successful extraction of known key bits or authentication outcomes from side channel data.
 - 4.2. Measure leakage using metrics like signal to noise ratio and correlation coefficients.



May–June: Complete project design plan and full hardware/software setup.
 Late May: Capture and analyse baseline power traces from RFID cards.
 Early June: Test encryption tools and choose a suitable cipher for experiments.
 Mid-June: Begin analysis of captured traces using statistical techniques.
 Late June: Continue capturing traces from additional card types and test cases.
 Late June – Early July: Develop automation scripts for trace handling and analysis.
 Mid-July: Finalise side channel attacks and complete results documentation.
 Late July: Complete writing of the final project report and materials.
 Early August: Conduct final validation of tools, testing, and results.
 Mid–Late August: Submit final project portfolio.
 Late August – Mid-September: Complete final interview and assessment.

SUCCESS CRITERIA

This project will be considered successful if the following criteria are met:

- Electromagnetic traces can be successfully captured.
- Side channel leakage correlating to key or encrypted value can be statistically demonstrated.
- Test data can be written and analysed on simple RFID memory cards.
- Trace analysis is reproducible using scripts or documented steps.
- The final report includes a clear summary of methods, findings, and ethical considerations.

This plan has been approved by my supervisor.

APPENDIX C

RESEARCH LOG

APPENDIX D

PROJECT DESIGN & IMPLEMENTATION

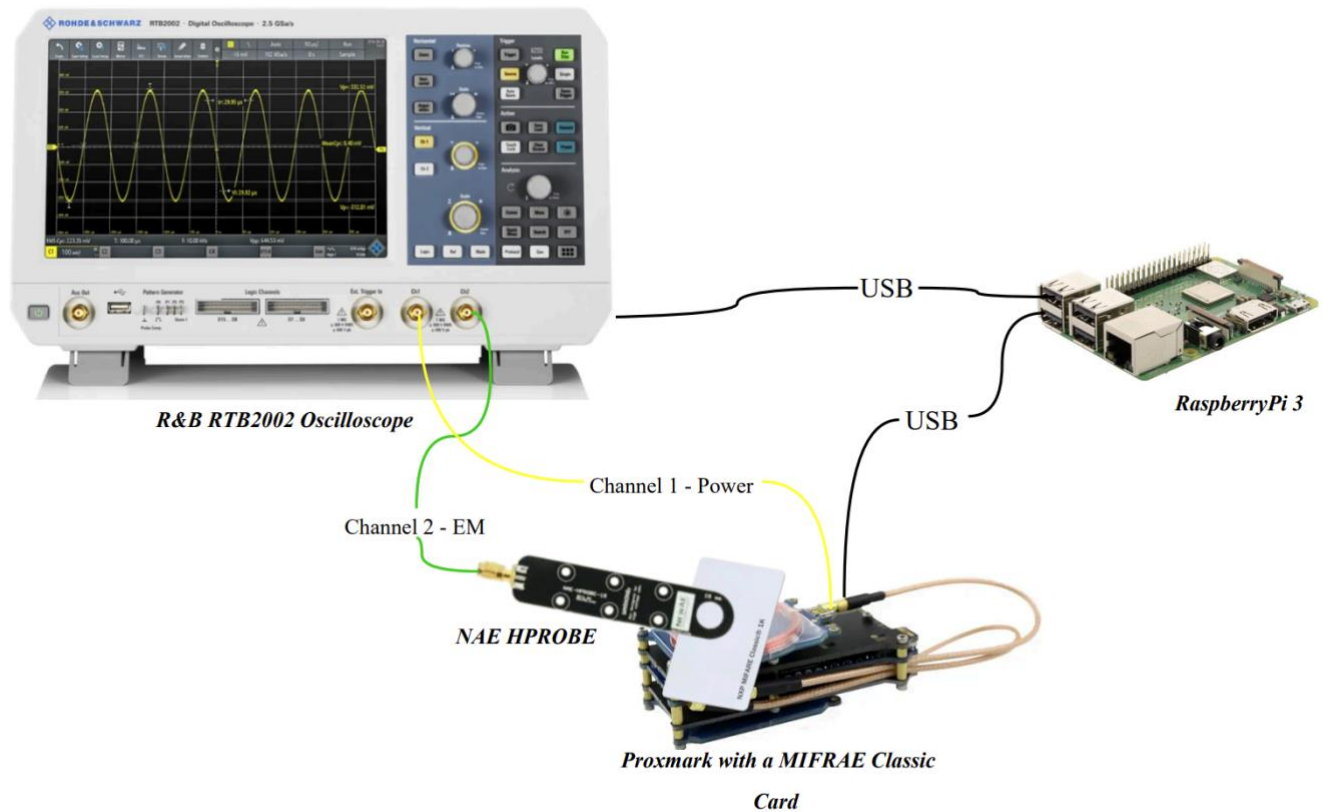
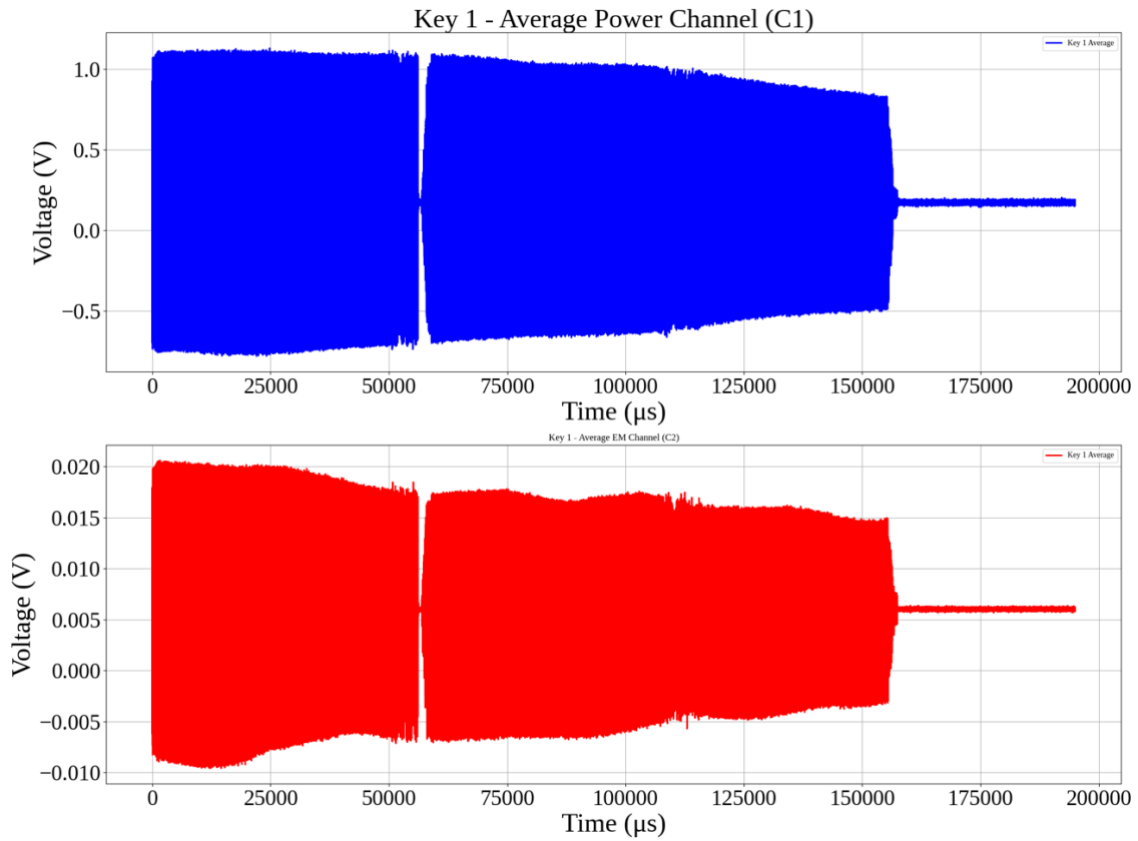


Figure 1: Experiment Hardware Setup

APPENDIX E

TESTING & RESULTS



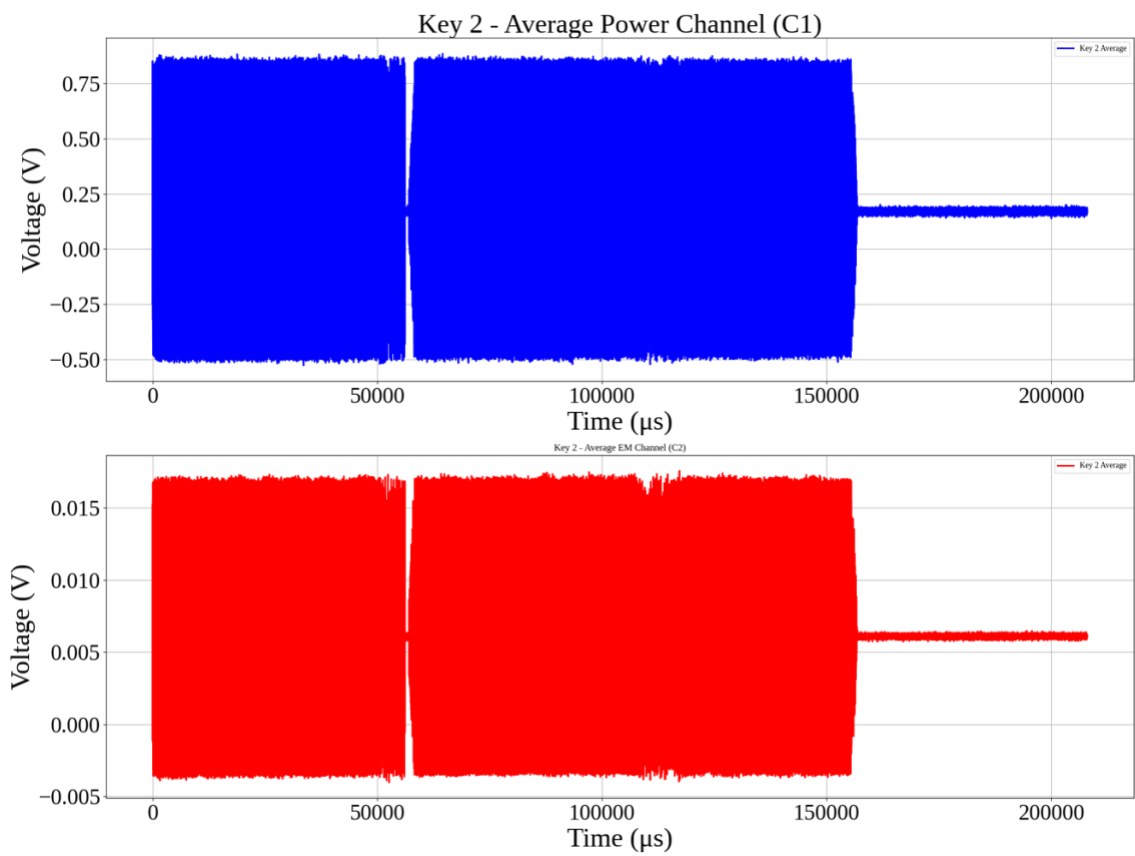


Figure 2:

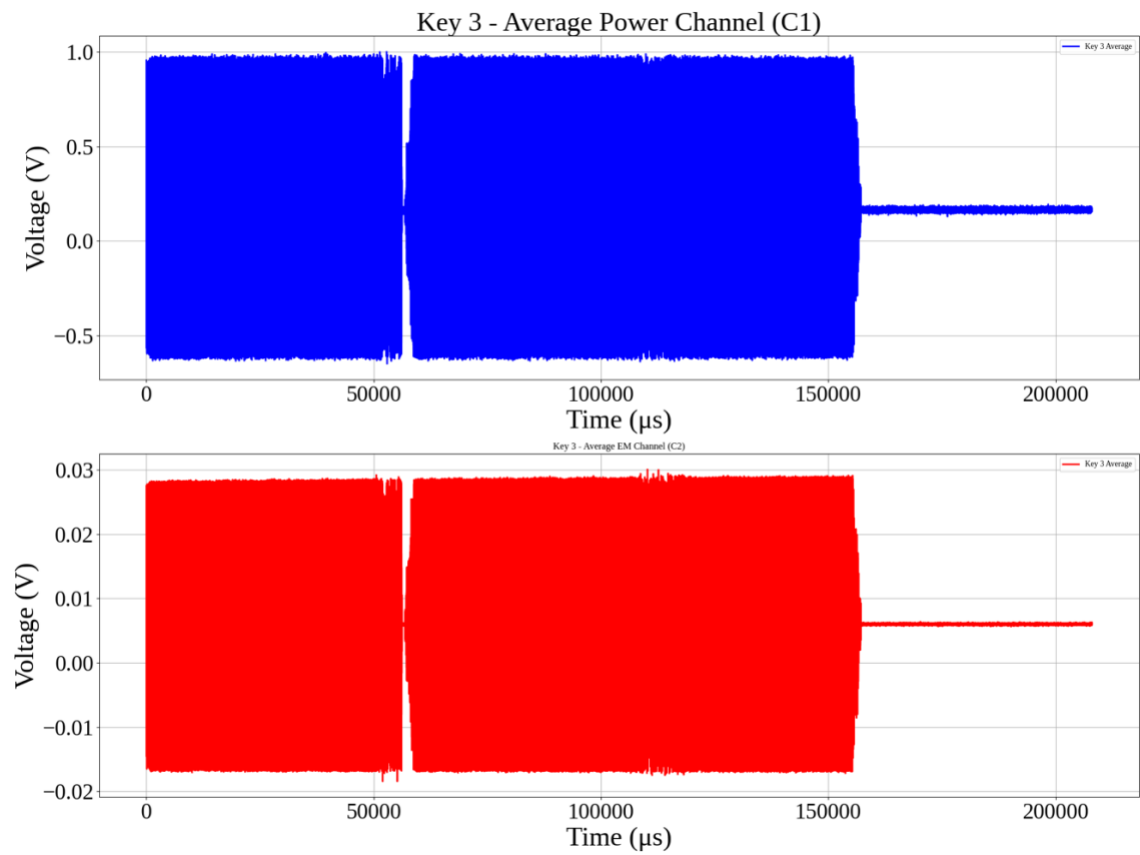


Figure 3:

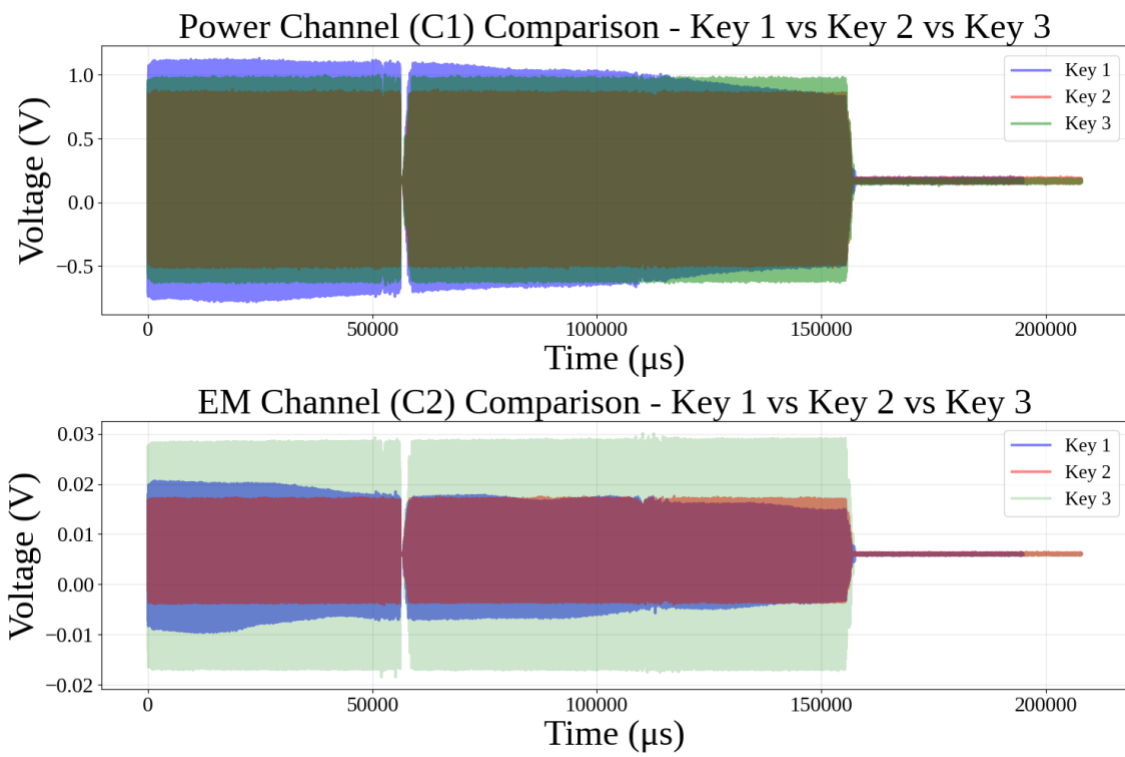


Figure 4:

APPENDIX F

SOURCE CODE AND CSV FILES

LINK TO GOOGLE COLAB

All the code used for preprocessing and analysis can be found in [this](#) Google Colab. Each section is clearly marked with headers, the code is fully commented, and all generated figures include titles and legends, ensuring that it is easy to follow and reproduce.

LINK TO GOOGLE DRIVE

All the unmodified CSV files that were downloaded from the oscilloscope can be found [here](#). These were originally named using the format CARD1_X, though a more accurate label would be BLOCK to reflect the specific card block tested. The reader should note this naming mismatch when interpreting the datasets.

APPENDIX G

MATHEMATICAL EQUATIONS

INDEPENDENT TWO-SAMPLE T-TEST

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

(1) [1]

- $\bar{X}_1$ and $\bar{X}_2$ are the sample means of groups 1 and 2
- s_1^2 and s_2^2 are the sample variances of groups 1 and 2
- n_1 and n_2 are the sample sizes of groups 1 and 2

the appropriate degrees of freedom were computed using Welch's formula to account for unequal group variances:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

(2) [1]

The independent t-test checks whether the difference between two groups is meaningful. Alongside the t-statistic, it produces a *p*-value. The *p*-value is the chance that the observed difference happened randomly. A *p*-value below 0.05 is typically considered “statistically significant,” meaning it's unlikely that the difference is due to chance alone.

COHEN'S *d*

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{pooled}}$$

(3) [2]

- M_1 and M_2 are the means of the two signals
- s_{pooled} is the pooled standard deviation:

$$s_{pooled} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

(4) [2]

- s_1 and s_2 are the standard deviation of each signal.

Values around 0.2 are considered small, 0.5 moderate, and above 0.8 large. Higher values indicate bigger differences between signals.

PEARSON CORRELATION COEFFICIENT

$$r = \frac{\sum (x - m_x)(y - m_y)}{\sqrt{\sum (x - m_x)^2 \sum (y - m_y)^2}}$$

(5)

[3]

- m_x and m_y are the mean of the vectors x and y

Values range from -1 to 1 . A value close to 0 means little or no relationship, values near 1 mean a strong positive relationship, and values near -1 mean a strong negative relationship. A value above 0.5 generally suggests a moderate to strong relationship.

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APPENDIX H

PROXMARK3 COMMAND LINE INTERFACE OUTPUT

[usb] pm3 --> hf search

🕒 Searching for ISO14443-A tag...

[=] ----- ISO14443-A Information -----

[+] UID: 6D 04 24 C4 (ONUID, re-used)

[+] ATQA: 00 04

[+] SAK: 08 [2]

[+] Possible types:

[+] MIFARE Classic 1K

[=]

[=] Proprietary non iso14443-4 card found

[=] RATS not supported

[#] Static nonce..... 01200145

[+] Static nonce..... yes

[?] Hint: Try `hf mf info`

[+] Valid ISO 14443-A tag found

[usb] pm3 --> hf mf info

[=] --- ISO14443-a Information -----

[+] UID: 6D 04 24 C4

[+] ATQA: 00 04

[+] SAK: 08 [1]

[=] --- Keys Information

[+] loaded 2 user keys
[+] loaded 61 hardcoded keys
[+] Sector 0 key A... 434A9734E087
[+] Sector 0 key B... FFFFFFFFFFFFFF
[+] Sector 1 key A... 010203040506
[+] Sector 1 key B... FFFFFFFFFFFFFF
[+] Block 0..... 6D0424C4890804006263646566676869 | bcdefghi

[=] --- Fingerprint

[+] Fudan based card

[=] --- Magic Tag Information

[=] <n/a>

[=] --- PRNG Information

[#] Static nonce..... 01200145

[+] Static nonce... yes

[usb] pm3 --> hf mf chk --1k -a --tblk 0 -k 434A9734E087

[+] loaded 1 user keys

[+] loaded 61 hardcoded keys

[=] Start check for keys...

[=] ..

[=] Time in checkkeys 0 seconds

[=] Testing to read key B...

[=] Sector: 0, First block: 0, Last block: 3, Num of blocks: 4

[=] Reading sector trailer

Data:FF FF FF FF FF FF

[+] -----+-----+-----+-----+-----+-----

[+] Sec | Blk | key A |res| key B |res|

[+] ----+----+-----+--+-----+----

[+] 000 | 003 | 434A9734E087 | 1 | FFFFFFFFFFFFFF | 1

[+] 001 | 007 | ----- | 0 | ----- | 0

[+] 002 | 011 | ----- | 0 | ----- | 0

[+] 003 | 015 | ----- | 0 | ----- | 0

[+] 004 | 019 | ----- | 0 | ----- | 0

[+] 005 | 023 | ----- | 0 | ----- | 0

[+] 006 | 027 | ----- | 0 | ----- | 0

[+] 007 | 031 | ----- | 0 | ----- | 0

[+] 008 | 035 | ----- | 0 | ----- | 0

[+] 009 | 039 | ----- | 0 | ----- | 0

[+] 010 | 043 | ----- | 0 | ----- | 0

[+] 011 | 047 | ----- | 0 | ----- | 0

[+] 012 | 051 | ----- | 0 | ----- | 0

[+] 013 | 055 | ----- | 0 | ----- | 0

[+] 014 | 059 | ----- | 0 | ----- | 0

[+] 015 | 063 | ----- | 0 | ----- | 0

[+] ----+----+-----+--+-----+----

[+] (0:Failed / 1:Success)